

■ 1-MINUTE VIDEO PRESENTATION GUIDE

AI-Based Loan Approval & Risk Assessment Platform (100% Complete)

1. PROJECT COMPLETION STATUS

Current Completion: 100% Fully Complete (Phases 1 through 5)

This represents the full end-to-end completion of **Phase 1 (Frontend UI)**, **Phase 2 (Database & REST API)**, **Phase 3 (XGBoost ML Pipeline)**, **Phase 4 (BI Analytics & Power BI Data Export)**, and **Phase 5 (Automated Pytest Testing & Security)**. The system is production-ready.

2. FILE STRUCTURE BREAKDOWN (2-LINE EXPLANATIONS)

- **train.py**: Generates the synthetic credit dataset, performs pre-screening logic, trains the baseline XGBoost classifier pipeline, and serializes the model to disk.
- **export_powerbi_dataset.py**: Queries the SQLite database logs, calculates BI dimensions (Risk Tiers, DTI Brackets), and exports datasets to Excel (.xlsx) and CSV (.csv) for Power BI.
- **tests/test_system.py**: Executes an automated pytest suite testing data preprocessing, model inference accuracy, API REST routes, database CRUD, and PDF generation.
- **requirements.txt**: Configures required Python packages for running Streamlit, FastAPI, SQLAlchemy, XGBoost, ReportLab, Plotly, Pytest, and OpenPyXL.
- **app/database.py**: Establishes the SQLAlchemy engine to initialize and connect to a local SQLite database (loan_system.db) and defines the application logs table structure.
- **app/main.py**: Runs the FastAPI server hosting REST API endpoints that handle application parsing, load the XGBoost pipeline, evaluate ML predictions, export BI data, and write logs.
- **app/report.py**: Programmatically designs and generates a styled PDF Credit Risk Memo containing applicant metrics, risk levels, and underwriting justifications for compliance audits.
- **app/frontend.py**: Launches the Streamlit dashboard providing page navigation, numerical input form widgets, Plotly BI analytics, PDF memo downloads, and model retraining triggers.

3. WORD-FOR-WORD PRESENTATION SCRIPT

Use this script while recording your screen. Speak at a calm, natural pace.

Time (Sec)	Screen Actions (What to Show)	Speech Script (What to Say)
0:00 - 0:10	Show the ■ Executive Dashboard page with its metrics and charts.	"Hello! Today, I am presenting the 100% completed AI Loan Approval and Risk Assessment platform. The system features a full production architecture combining interactive Streamlit dashboards, FastAPI microservices, and automated testing."

0:10 - 0:25	Navigate to ■ Loan Application Form , enter profile details, and click Evaluate.	"Our application form captures applicant profiles and queries our trained XGBoost machine learning model. It calculates default risk probabilities, assigns risk tiers, and outputs the primary risk factors driving the recommendation."
0:25 - 0:45	Switch to ■ BI & Portfolio Analytics . Show Plotly scatter plots & click Export Power BI.	"In our Business Intelligence view, executives can inspect FICO scatter plots, portfolio distribution charts, and XGBoost feature importance weights. We can also export structured datasets directly to Excel and CSV for Power BI Desktop ingestion."
0:45 - 1:00	Go to ■ Historical Audit Logs , download a PDF Memo, and mention Pytest tests.	"Finally, historical records are logged to SQLite, and auditors can export a formal PDF Credit Memo. With automated pytest suites covering our APIs and models, the platform is 100% complete and production ready. Thank you!"